

Ravi Mukkapati

Data Architect | AWS Architect | Cloud Data Engineer

<https://psravi.github.io/>

A technically astute and competent individual with over 15 years of experience in the field of Cloud Engineering, Analytics, and Data Architecture, specializing in Healthcare, Pharma, and financial sectors. Seeking career advancement with an esteemed organization to contribute my skills in architecture, agile project management, and cloud service models (SaaS, PaaS, IaaS) to deliver transformative solutions that enhance operational excellence, compliance, and customer satisfaction whilst bridging the gap between technology and business needs, ultimately ensuring the success and growth of organizations in the healthcare, pharma, and financial sectors.

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SKILLS

Data Architecture

Database Management System (DBMS)

Data Engineering

Agile Development

Data Analytics

Software Project Management

DevOps Engineering

Requirements Analysis and Gathering

Business Intelligence

Data Governance and Quality

Application Development

AI & Machine Learning

Digital and Cloud Transformation

TECHNICAL SKILLS

Cloud Technologies: **AWS:** IAM, EC2, ECS, Lambda, EMR, S3, Glue, Athena, DMS, SageMaker, Step Functions, SQS, SNS
GCP: GCE, Cloud Storage, Dataflow, Dataproc/Spark, Cloud Functions, Big Query and Vertex AI.
OCI: Object Storage, Data Integration, OCI Data Flow, Functions, OCI Data Science and Azure DevOps.

Databases: Snowflake, Oracle ADW, ATP, MS SQL, MySQL, PostgreSQL, Mongo, Duck, Cache, KB, and Hadoop Hive.

OS Configuration: Unix, Linux Distributions, Windows, and Docker containerization
Terraform, CloudFormation

Programming: Python, PySpark, Shell, Java, SQL, PL/SQL, Apache Sqoop, Scala and HiveQL.

APIs / Web: FastAPI, Postman API Development, XML, XSLT, HTML, CSS, JSON, YML and Java Script

Oracle/PeopleSoft People Tools: HCM (Core HR, Benefits, NA Payroll and Employee/Manager Self-Service) ELM (Learning Management with internal/external content) CRM (Order capture, Sales, Support, Service and Help Desk) BI Publisher, Integration Broker, RESTful APIs and PS-Administration.

BI Tools: Oracle BI Apps, OBIEE 11g, DAC 10.x, Quick Sight, Tableau and Power BI.

Reporting Tools: SQR, BI Publisher, Crystal, PS-Query, n-Vision and Streamlit Apps

ETL/ELT/CDC: DBT Core, Fivetran, Prefect, Informatica, Oracle Goldengate, ODI, AWS DMS and Modelling Tools (Erwin, SqlDBM).

DevOps: GitHub, GitLab, Jira, Jenkins, Ansible, Bitbucket, SQL Liquibase, Confluence, ServiceNow, Scrum/ CI/CD
Apache Airflow and BMC Control-M.

AI/Machine Learning: Sentiment Analysis, Feature Engineering, ANN, CNN, RNN, TensorFlow, Keras, Scikit-Learn, Spark ML, Confusion matrix-based model evaluation, Linear & Logistic regression, Decision trees and pruning, Hyperparameter tuning, K-means Clustering, Hierarchical Clustering, Bagging Classifiers - Bagging and Random Forest, Boosting Classifiers -AdaBoost, Gradient Boosting.

CERTIFICATES

AWS Certified [Solutions Architect](#)

AWS Certified [Cloud Practitioner](#)

Snowflake Hands-On badges and [SnowPro](#) Certified

Oracle Certified Expert

PG Program in [AIML](#) from Texas University at Austin

People Tools Application Developer Certified Expert.

EDUCATION

AIML Post Graduate Program - University of Texas at Austin 2023 – 2024

Master's degree in computers Bharathidasan University 2000 – 2003

KEY STRENGTHS

Healthcare, Pharma, & Financial Industry Expertise:

Possessing in-depth knowledge of the specific regulatory frameworks, compliance requirements, and industry nuances within healthcare, pharma, and financial sectors, enabling the design and implementation of technology solutions that align seamlessly with industry standards and best practices.

Digital and Cloud Transformation:

Excelling in spearheading the migration and transformation of traditional on-premises infrastructure and applications into dynamic and scalable cloud environments, enabling organizations in healthcare, pharma, and financial sectors to harness the full potential of cloud technologies for enhanced agility, cost-efficiency, and innovation.

Enterprise Resource Planning:

Demonstrating expertise in optimizing the selection, deployment, and management of ERP systems tailored to the intricacies of healthcare, pharma, and financial sectors, enabling seamless data flow, process automation, and resource allocation across the entire organization.

Data Architecture:

Expertly designing data architectures that cater to the unique data needs of healthcare, pharma, and financial organizations, ensuring data security, integrity, and accessibility while enabling advanced analytics and informed decision-making to drive business success.

Enterprise Solution Architecture:

Proficiently crafting comprehensive and integrated architectural blueprints tailored to the unique requirements of healthcare, pharma, and financial industries, ensuring that all technology components align cohesively to support business objectives, security, and regulatory compliance.

Service-Oriented Architecture:

Skillfully designing and implementing service-oriented frameworks that facilitate modular and interoperable software components, promoting agility, scalability, and reusability while catering to the specific needs of healthcare, pharma, and financial domains.

WORK EXPERIENCE

Cloud Data Engineer Lead Baptist Health South Florida

02/2016 - Present,

Growth Path: Application Architect (02/2016-08/2019) | Cloud Data Architect (08/2019- Present)

• **Project Description:** Cloud Data Migration and Modernization focused on migrating enterprise healthcare data platforms from on-prem Hadoop Hive and Oracle environments to AWS and Snowflake. As a Cloud Data Engineer Lead, responsible for designing scalable data pipelines, automation frameworks, Airflow orchestration, CI/CD deployment, and modern ELT solutions for reporting, analytics, and downstream data consumption.

- Led the migration of enterprise healthcare data platforms from on-prem Hadoop Hive and Oracle environments to AWS and Snowflake, supporting scalable analytics and a modern cloud-based ELT architecture.
- Built reusable ingestion and transformation frameworks using Python, Spark, DBT, and Airflow to process API-based, file-based, and database-sourced data across Snowflake raw, stage, and EDW layers.
- Developed and orchestrated Airflow DAGs to automate data movement from source systems to AWS S3 and Snowflake, including historical loads, incremental loads, soft deletes, merges, validations, error handling, and rerun processes.
- Implemented CI/CD pipelines using GitLab and Terraform to automate the deployment of data pipelines, Snowflake objects, infrastructure components, and AWS services.
- Built API ingestion frameworks using Python, FastAPI, Postman, and Spark to integrate enterprise and third-party systems such as Salesforce, ServiceNow, Zoom, Credential Stream, and Midas into the cloud data ecosystem.
- Supported AWS to GCP migration activities by assisting with data platform assessment, pipeline migration planning, cloud service mapping, and validation of data movement across cloud environments.
- Improved data quality, governance, and reliability across healthcare datasets by implementing validation rules, reconciliation checks, automated monitoring, and standardized ETL/ELT patterns for downstream reporting, analytics, and data consumption.
- Architecting Technical Solutions: Designing technical solutions for both new and existing processes to address complex business challenges and facilitate automation by leveraging a diverse array of tools.
- Implemented end-to-end data solutions for Provider Data Management, resulting in a significant reduction in data errors and improved data quality.

PAST EMPLOYMENT HISTORY

Baptist Health South Florida as ERP Application Architect (2016 - 2019)
JP Morgan Chase as Application Development Lead (2013 - 2015)
Int Info Tech as Oracle consultant (2009 - 2012)
8x8 Inc as Technical Lead (2008 - 2009)
IBM Pune, India as Project Lead (2006 - 2007)
TCS Bangalore as Software Engineer (2004 - 2005)

PROJECTS

Project Title: Enterprise Application Architecture and Integration | Client Name: Baptist Health | Role: Application Architect (02/2016 - 07/2019)

- **Project Description:** Implemented Oracle BI Apps and a comprehensive suite of Oracle ERP modules (HCM, ELM, Portal, CRM, and FSCM) for Baptist Health, a healthcare organization with multiple hospitals, primary care, and urgent care locations across South Florida. The project involved migrating data from Lawson to the new PS ERP application.
- **Project Responsibilities:** Gathered requirements from various business stakeholders, including HRIS, Payroll, Benefits, and BI, and translated them into technical specifications.
- Architected reliable solutions and managed the end-to-end development, testing, and deployment processes.
- Executed critical projects, including Employee Payroll integrations, Benefit integrations, ELM integrations, data migrations during hospital acquisitions, and vendor integrations using files, XML, and Restful APIs.
- Developed Employee and Manager self-service fluid web pages, conducted SQL query performance tuning, and implemented data archival and data masking.
- Designed and developed BI reports and dashboards using AWS services such as S3, Glue, and Athena, and implemented CI/CD workflows using Apache Airflow DAGs. Provided support for multiple PeopleSoft modules, including Core HR, Payroll, Benefits, ESS/MSS, CRM, and ELM, and actively participated in patching and upgrade projects.
- Conducted design and code reviews, ensuring adherence to quality, security, and requirements. Led technical presentations and mentored team members in troubleshooting and technical problem-solving.
- Implemented data safeguarding projects, including Oracle data masking, encryption, and sensitive data hiding to mitigate PII exposure risk. Developed custom BI Reports for Employee self-service, such as Employment Verification, Merit, Salary, and Benefit confirmation letters.
- Conducted comprehensive data analysis and recommended solutions for complex business problems using advanced analytical methods, including the development of database objects with SQL.
- Provided support for the Enterprise Learning Module (ELM) by creating custom components and fluid pages to meet BHSF's specific business needs, including custom admin features, course permissions, course completion certificates, reports, and external user enrollment and payment capabilities for clinical courses offered through AHA and other programs.

Project Title: Payment Processing Application Development | Client Name: JPMorgan Chase & Co (Chase Paymentech) | Role: Application Software Development Lead | 05/2013 - 01/2016 (09/2009 - 03/2011)

- **Project Description:** Developed payment processing applications for JPMorgan Chase, encompassing credit/debit/gift/pre-paid and ACH payments online and at point-of-sale (POS). Collaborated on end-to-end merchant services for Chase Paymentech, while also maintaining the ERP system managing critical merchant and customer data.
- **Project Responsibilities:** Led cross-functional teams in designing and developing payment processing applications. Collaborated with third-party vendors, internal teams, and business units to shape roadmaps and resolve issues. Developed and executed comprehensive strategies, focusing on web services and payment integration.
- Participated in Business Requirement Analysis meetings, contributing to functional documentation.
- Mentored the team in integration tools, coordinated efforts, assigned tasks, and conducted code reviews. Led projects involving batch processing, data security, data archival, and more.
- Worked extensively with integration tools, transforming XML data and implementing SOA web services.
- Created Configurable Search components and managed field-level permissions for data protection.
- Developed Shell Scripts for secure file transmission and encryption.
- Designed and scheduled jobs for archiving large tables efficiently. Enhanced Oracle Database objects and procedures for improved performance.
- **Environment :** CRM with Tools 8.51, Oracle Database, Application Designer, Git, CONTROL-M, and more.

Project Title: Data Warehouse Enhancements and Support | Client Name: Asplundh Tree Expert (I2t2 Inc) | Role: Data Analytics Consultant (12/2012 - 04/2013)

- **Project Description:** Worked as a Senior Consultant to enhance and support the Data Warehouse (DWH) for Asplundh, focusing on Equipment Management components and Crystal Reports.
- **Project Responsibilities:** Extensively worked on enhancing Equipment Management components to align with Asplundh's business model.
- Created and modified multiple Crystal Reports, including Asset Depreciation, Disposal, Asset by State, and Asset Transfer Reports in the Asset Management module.
- Modified the Vouchers Interface to accommodate FedEx and Telecom data, creating a custom report to track this data.
- Ensured accuracy in the Invoices report, specifically correcting Original Invoice and Final values for vendors with missing contacts.
- Developed and enhanced various App Engine programs to facilitate data loading from external systems into PeopleSoft.
- Created and scheduled OBIEE programs to load and analyze Procurement and Spend data.
- Developed multiple OBIEE reports, providing insights into PeopleSoft data, such as Expense Reports by Business Unit.
- **Environment:** Oracle PeopleSoft FSCM 9.1 with Tools 8.51, Oracle Database, and BI Tools including PeopleSoft to OBIEE Procurement and Spend Analysis Dashboards and Reports.

PROJECTS

Project Title: Data Warehouse Implementation and Business Intelligence Support | Client Name: Affymax/Takeda Pharmaceuticals (I2t2 Inc) | Role: Data Analytics/BI Consultant (03/2011 - 11/2012)

- **Project Description:** Worked as a Consultant on the implementation and support of the Data Warehouse (DWH) for Affymax/Takeda Pharmaceuticals, involving extensive business requirements gathering and report design.
- **Project Responsibilities:** Led business requirements gathering meetings with Executive Management and business users. Translated requirements into Functional and Technical specification documents.
- Designed mock-up reports and negotiated with users on report requirements. Created OBIEE 11g environments with necessary configurations, including LDAP, Maps, Office, and usage Analytics. Utilized Informatica for data loading from multiple sources (Takeda, IMS Drug Distribution Data, Salesforce, and Call Data) into the Data Warehouse.
- Designed data flow processes using Informatica client tools, including Source Analyzer, Warehouse designer, Mapping Designer, Mapplet Designer, and Transformation Developer. Developed web services and integrated them within OBIEE for enhanced functionality. Configured Navteq Maps to visualize ZIP Level Sales Data and implemented Data Level security to restrict user access.
- Developed PL/SQL procedures and functions in Informatica to optimize performance. Designed dimension hierarchies, aggregates, and logical sources in the BMM layer for multidimensional data in OBIEE dashboard reports.
- Created various types of reports and dashboards, including Drill-Down/Dynamic, Pivot Tables, and Column Selector.
- Integrated Salesforce Veeva Application with OBIEE to provide real-time Account, Contact, and call data in analysis reports.
- **Environment:** Oracle BI Applications 7.9.x Analytics, Salesforce Veeva, OBIEE 11.1x, Informatica 9, Oracle 11g, Oracle Data Warehouse Administration Console (DAC) 10.1.3.

Project Title: CRM and Order Capture Module Enhancement | Client Name: 8x8 Inc | Role: Technical Lead (08/2007 - 04/2009)

- **Project Description:** Worked as a Senior Software Engineer on enhancements and support for the CRM and Order Capture modules at 8x8 Inc, a provider of VoIP-based SIP trunking phone services.
- **Project Responsibilities:** Developed APIs for integration with internal and external applications, including Level3 provider and shippers.
- Functioned as a CRM Techno-Functional consultant and contributed to the design and development of modules supporting the Yahsat business unit.
- Implemented synchronous APIs for real-time data synchronization with other systems like DSS, FAP, and Finance.
- Configured various components such as 360-Degree view, Tree Nodes, Profiles, Roles, and search preferences.
- Created Configurable search Pages and set up components to enforce data-level security.
- Conducted customizations and worked extensively on components related to Site, 360-View, Company, Contacts, Workers, Provider Groups, and Support Case management.
- **Environment:** CRM 8.9 with PeopleTools 8.49, Oracle, Salesforce, PeopleSoft 8.9 with Tools 8.4x, Oracle, SQL Server Databases, Integration Broker, XML, and XSLT.

Project Title: HCM and CRM Enhancements and Support | Client Name: Panasonic Electronics (IBM India/USA) | Role: Project Lead (06/2005 - 07/2007)

- **Project Description:** Worked as a Senior Consultant on the enhancement and production support activities for PeopleSoft HCM and CRM used by Panasonic Electronics.
- **Project Responsibilities:** Provided extensive support for HCM and CRM components, including Support, Case, and 360-degree views.
- Developed PS-Queries, SQR, and ad-hoc reports for web and call center cases.
- Customized various delivered components, such as Agreements, warranties, Support, Order entry, and product management.
- Designed and implemented the Product registration process.
- Developed synchronous and asynchronous integrations with internal and external systems using Integration Broker objects and XML/XSLTs.
- **Environment:** PeopleSoft CRM and HCM 8.8 with Tools 8.4x on Oracle Database.

Project Title: CRM Application Development and Customization | Client Name: Apple Inc (TCS) | Role: Software Engineer (04/2003 - 05/2005)

- **Project Description:** Worked as a Software Engineer in the CRM project at Apple Inc, where I participated in all phases of application development, from requirements gathering to deployment.
- **Project Responsibilities:** Played a key role in the migration of Apple's Vantive CRM application to the PeopleSoft ERP system.
- Developed and customized various PeopleSoft CRM components and pages tailored to the Apple Asia Pacific region, including customer Self Service and Agreement pages in multiple localized languages for APAC and European countries.

PERSONAL ATTRIBUTES

Intellectual Curiosity

Impeccable Analytical skills

Innate Problem-solver

Technology Enthusiastic

Team Player

LANGUAGES

English
Full Professional Proficiency

Hindi
Full Professional Proficiency

Telugu
Full Professional Proficiency

Tamil
Full Professional Proficiency

INTERESTS



Yoga



Charity/Community
Service



Travelling



Mentoring